Content:

Quantum Mechanics: de Broglie Hypothesis and Matter waves, Heisenberg's uncertainty principle and its application in broadening of spectral lines.

Wave Mechanics: Wave function and its properties, Operators, Eigen functions and Eigen values of an operator, Example of a free particle, Expectation value and its significance, Derivation of 1-dimensional (1D) time-independent Schrödinger equation (TISE) from classical wave equation, Solving the 1D TISE for infinite well, Recovery of free particle solution for infinite well width, Solution for 2 and 3-dimensional (3D) well by analogy with 1D, Degeneracy in higher dimensions, Finite well (concept of symmetry) and Quantum Tunnelling, Numerical problems.

Introduction:

At the beginning of the 20th century, Newton's laws of motion were able to successfully describe the motion of the particles in classical mechanics (the world of large, heavy and slow bodies) and Maxwell's equations explained phenomena in classical electromagnetism. However, the classical theory does not hold in the region of atomic dimensions. It could not explain the stability of atoms, energy distribution in the black body radiation spectrum, origin of discrete spectra of atoms, etc. It also fails to explain the large number of observed phenomena like photoelectric effect, Compton Effect, Raman Effect, Quantum Hall effect, superconductivity etc. **The insufficiency of classical mechanics led to the development of quantum mechanics (QM).**

Quantum mechanics gives the description of motion and interaction of particles in the small-scale atomic system where the discrete nature of the physical world becomes important. With the

application of quantum mechanics, most of the outstanding problems have been solved. The direct implications of QM though very subtle have dramatic effect in our day-to-day life. For example, QM successfully introduced the concept of discrete energy which led to the conception and advancements in quantum computing, it also paved way for efficient long-distance transfer of large quantities of data electronically. By understanding the electronic spin and related atomic properties, QM has aided in realizing energy efficient materials that can be applied to consumable electronic equipment and in commercial transport vehicles like Hyper loop, Maglev trains among others.

Wave and particle duality of radiation

To understand the wave and particle duality, it is necessary to know what a particle is and what a wave is.

A particle is a localized mass and it is specified by its mass, velocity, momentum, energy, etc. In contrast a wave is a spread-out disturbance. A wave is characterized by its wavelength, frequency, velocity, amplitude, intensity, etc. It is hard to think mass being associated with a wave. Considering the above facts, it appears difficult to accept the conflicting ideas that radiation has a wave particle duality. However, this acceptance is essential because the radiation exhibits phenomena like interference, diffraction, polarization, etc., and shows the wave nature and it also exhibits the particle nature in the black-body radiation effect, photoelectric effect, Compton Effect etc.

Radiation, thus, sometimes behave as a *wave* and at some other time as a *particle*, this is the wave particle duality of radiation.

de Broglie's concept of matter waves.

Louis de-Broglie in 1924 extended the wave particle dualism of radiation to fundamental entities, such as electrons, protons, neutrons, atoms, molecules, etc. de-Broglie put a bold suggestion that like radiation, matter also has dual characteristic, at a time when there was absolutely no experimental evidence for wavelike properties of matter waves. **de-Broglie Hypothesis** of matter waves is as follows.

- In nature energy manifests itself in two forms, namely matter and radiation.
- Nature loves symmetry.
- As radiation can act like both wave and a particle, material particles (like electrons, protons, etc.) in motion should exhibit the property of waves.

These waves due to moving matter are called *matter waves or de-Broglie waves or pilot waves.*

Wavelength of matter waves:

The concept of matter waves is well understood by combining Planck's quantum theory and Einstein's theory. Consider a photon of energy E, frequency γ and wavelength λ.

By Planck's theory $E = h\gamma = \frac{hc}{\lambda}$

By Einstein's mass-energy relation $E = mc^2$

By equating and rearranging the above equations, we get

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

Where, p is the momentum of the photon and h is a Planck's constant.

Now consider a particle of mass m moving with a velocity v and momentum p. According to the de-Broglie hypothesis matter also has a dual nature. Hence the wavelength λ of matter waves is given by

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

This is the equation for the de-Broglie wavelength.

de Broglie wavelength of an electron

Consider an electron of mass 'm' accelerated from rest by an electric potential V. The electrical work done (eV) is equal to the kinetic energy E gained by the electron.

$$E = eV$$
$$\therefore E = \frac{1}{2}mv^2$$
$$\therefore m^2v^2 = 2mE$$
$$mv = \sqrt{2mE}$$

Therefore, wavelength of electron wave

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mE}}$$

But $E = eV$

$$\therefore \lambda = \frac{h}{\sqrt{2meV}} = \frac{\frac{h}{\sqrt{2me}}}{\sqrt{V}}$$

Substituting the values of e, m and h, we get